

RASAero II Comparisons with Altitude Data

The accuracy of the RASAero II flight simulation for high power rockets and amateur rockets is shown in the Plot and Table below, where the RASAero II flight simulation apogee altitude is compared with barometric altimeter data, optical tracking data, accelerometer-based flight data, and GPS flight data for a representative set of high power rockets and amateur rockets. RASAero II rocket files (.CDX1 files) for most of these rockets are included in the RASAero II software download.

All altitudes are Above Ground Level (AGL).



Comparison of RASAero II Flight Simulation Apogee Altitude with Barometric Altimeter, Optical Tracking, Accelerometer-Based, and GPS Apogee Altitude Flight Data

Rocket	Motor(s)	Diameter (in)	Flight Data	Apogee Altitude (ft)	RASAero Flight Sim Apogee Altitude (ft)	Percent Error
MESOS 293K ft Flight	O4374 (KIP-EX) M787 (KIP-EX)	Booster = 4.68 Sustainer = 3.15	GPS (See Note 1)	293488	289789	-1.26%
ThreeCarbYen 2018	N5800 (CTI) N1560 (CTI) M1401 (CTI)	Stage-1 = 4.28 Stage-2 = 4.1 Stage-3 = 3.17	GPS (See Note 2)	175000	185018	5.72%
AeroPac 104K ft Two-Stage Rocket	N1048 (AT) M685W (AT)	Booster = 4.05 Sustainer = 3.08	GPS (See Note 3)	104659	113786	8.72%
Qu8k	Q18000 (Deville-EX)	8.000	Integrated Accelerometer Data (See Note 4)	121478	116254	-4.30%
Our Project R Rocket	R17971	10.500	Time to Apogee from Onboard Video (See Note 5)	94000	102402	8.94%
Proteus 6	P9381 (Loki-EX)	6.000	Integrated Accelerometer Data (See Note 6)	85067	86799	2.04%
Proteus 6.5	P9381 (Loki-EX)	6.000	Average of GPS and Integrated	72233	76639	6.10%

			Accelerometer Data			
Don't Debate This	N5800 (CTI)	3.9173	Barometric Altimeter (See Note 7)	56573	62308	10.14%
A-601	P4935	6.125	GPS (See Note 8)	42771	41086	-3.94%
Full Metal Jacket - BALLS 005	O10000 (Kosdon)	4.000	Optical Track	37981	38820	2.21%
Full Metal Jacket - Black Rock-6	O10000 (Kosdon)	4.000	Optical Track	30038	32646	8.68%
Kline-Rogers L500 Rocket	L500 (Ace)	2.260	Optical Track	24771	26485	6.92%
Violent Agreement SS Sustainer	J530 (CTI)	1.600	Barometric Altimeter (See Note 9)	22874	25602	11.93%
Violent Agreement	I216 (CTI)	1.600	Barometric Altimeter (See Note 9)	18495	19256	4.11%
Laser/LOC-2.1	J125	2.260	Optical Track	15818	14868	-6.01%
Laser/LOC-2.1	J125	2.260	Optical Track	13219	14616	10.57%
Torrent	M1850GG (AMW)	4.000	Barometric Altimeter	12807	13852	8.16%
Rabia	L1080BB (AMW)	3.000	Barometric Altimeter	12745	12777	0.25%
Hubbub	L1080BB (AMW)	4.024	Barometric Altimeter	10750	10883	1.24%
Rabia - Short Fin Can	L730 (CTI)	3.126	Barometric Altimeter	10584	10376	-1.97%
Blister	K1075GG (AMW)	3.000	Barometric Altimeter	9026	8347	-7.52%
Raven	J570W (AT)	1.750	Barometric Altimeter	8815	9288	5.37%
EZI-65 J125	J125	3.998	Optical Track	8068	7436	-7.83%
Ion Drive	K550W (AT)	4.000	Barometric Altimeter	8027	8642	7.66%
Cancer Descending	M1297W (AT)	6.000	Barometric Altimeter	6188	6328	2.26%
Byrum	J570W (AT)	3.000	Barometric Altimeter	5732	5280	-7.89%
EZI-65 J100	J100	3.998	Optical Track	5671	6472	14.12%
Caliber Isp 05 ARO-414	I285 (CTI)	3.100	Barometric Altimeter	5085	4842	-4.78%
Caliber Isp 05 ARO-414	I285 (CTI)	3.100	Barometric Altimeter	4930	4831	-2.01%

McGarvey	J350W-L (AT)	3.000	Barometric Altimeter	4246	4862	14.51%
EZI-65	J450ST (AMW)	4.000	Barometric Altimeter	3965	4214	6.28%
Caliber Isp 04 AVTC Team 3	I205 (CTI)	3.100	Barometric Altimeter	3964	3871	-2.35%
Gibb	I284W (AT)	3.000	Barometric Altimeter	3913	4310	10.15%
Caliber Isp 04 AVTC Team 1	I205 (CTI)	3.100	Barometric Altimeter	3837	3943	2.76%
Caliber Isp 04 AVTC Team 2	I205 (CTI)	3.100	Barometric Altimeter	3710	3871	4.34%
Thunder & Lightning	I284W (AT)	3.100	Barometric Altimeter	3577	3989	11.52%

Average
Error = 3.47%

80.6% of Flights
Error < 10%

41.7% of Flights
Error < 5%

Note 1: Altitude based on GPS data. Rocket carried a Kate onboard instrumentation unit. Accelerometer-based altitude and barometric altitude also obtained during the flight.

Data from Kate Instrumentation Unit:

All Altitudes are Above Ground Level (AGL)

GPS Apogee Altitude = 293,488 ft
Accelerometer-Based Altitude = 294,271 ft
Barometric Altimeter Apogee Altitude = 275,131 ft

GPS Downrange Distance at Apogee = 55,767 ft

From Accelerometer data rocket reached 4,047 ft/sec (Mach 4.18) at 62,301 ft

RASAero II Flt Sim Apogee Altitude is a postflight simulation where Sustainer (Stage-2) ignition delay time has been adjusted to match the flight data, and the launch angle from vertical has been adjusted to match the downrange distance at apogee.

Note 2: Altitude based on GPS data. Last GPS data point on ascent before predicted apogee was at 160,000 ft. Next GPS data point was received after predicted apogee during descent, and was just under 160,000 ft. From extrapolation of GPS altitude data with time estimated apogee altitude was 175,000 ft.

Note 3: Altitude based on GPS data. GPS Downrange Distance at Apogee = 17.1 miles (statute miles, US miles). RASAero II Flt Sim Apogee Altitude is a postflight simulation where the launch angle from vertical has been adjusted to match the downrange distance at apogee.

Note 4: Altitude based on integration of onboard accelerometer data. Altitude based on the average of the apogee altitude from the two onboard accelerometers. The rocket had a very vertical flight, and landed only 3 miles from the launch site.

Note 5: Altitude based on time to apogee from onboard video (Reference: Postflight Analysis of the OuR Project R Rocket Flight, *High Power Rocketry* magazine, July 1997). From onboard video rocket spin rate 0.26-0.37 revolutions/sec. RASAero II flight simulation based on 3.5 degree coning angle from onboard video.

Note 6: Altitude based on integration of onboard accelerometer data. Altitude based on the average of the apogee altitude from the two onboard accelerometers. Rocket also carried a barometric altimeter and a GPS unit. Data from barometric altimeter and GPS unit stopped above 46000 ft. Integrated accelerometer data agreed well with barometric altimeter data up to 46000 ft, barometric altimeter data agreed well with GPS data below 46000 ft during parachute descent.

Note 7: Altitude based on the average of the apogee altitude from Featherweight and PerfectFlite barometric altimeters. Featherweight barometric altimeter altitude was 56504 ft, PerfectFlite barometric altimeter altitude was 56641 ft.

Note 8: Altitude based on GPS data. Rocket also carried two barometric altimeters and an accelerometer. Altitudes from the different onboard instrumentation were the following:

GPS	42,771 ft
Adept	42,231 ft (barometric altitude)
ARTS II	40,113 ft (barometric altitude)
ARTS II	44,924 ft (integrated accelerometer altitude)

The rocket had a very vertical flight, and landed only 2 miles from the launch site.

Note 9: Altitude from barometric altimeter, with the altitude data adjusted based on balloon-referenced atmospheric pressure data, from a balloon launched prior to flight.

As can be seen in the Plot and the Table above, for this representative set of high power rocket and amateur rocket barometric altimeter data, optical tracking data, accelerometer-based flight data, and GPS flight data, the average error in the RASAero II flight simulation apogee altitude was 3.47%, with 80.6% of the flights within +/-10%, and 41.7% of the flights within +/-5%.